Supplementary information: compressing dense similarity for dataset splitting

Author information pending

1 Dataset scope

Target slices follow the declared dataset-loader mapping and supply one scalar task per source for the prediction studies. Structural benchmarks optimize features without using responses, including entities whose target is missing. The study does not claim coverage of every target in Tox21, SIDER or MUV. MUV-466 has sparse labels and few actives; no predictive AUC is reported for MUV in this article. The dataset catalogue identifies the exact offline snapshots rather than silently substituting a newer database query.

Dataset	Entities	Labelled	Target	Positive
bace	1513	1513	Class	691
bbbp	2039	2039	p_np	1560
esol	1128	1128	measured log solubility in mols per litre	–
freesolv	642	642	expt	–
lipophilicity	4200	4200	exp	–
clintox	1480	1480	CT_TOX	112
hiv	41120	41120	HIV_active	1443
tox21	7823	7258	NR-AR	308
sider	1427	1427	Hepatobiliary disorders	743
muv	93087	14841	MUV-466	27
qm8	21786	21786	E1-CC2	–
qmof	20372	20372	outputs.pbe.bandgap	–
materials project	100000	99998	formation_energy_per_atom	–

Table 1: Retained entities, declared target slices and label coverage. Positive counts apply only to binary tasks; dashes denote continuous targets. Structural splitting retains valid features independently of missing responses. Single-target slices do not constitute an evaluation of all assays in a multitask source. Exact raw-file and retained-identity hashes accompany the catalogue.

Setting	Power	Group threshold	Capacity tolerance
Plain	1	None	0, 0.01 or 0.05
Power only	3	None	0
Groups only	1	0.4	0
Combined	3	0.4	0, 0.01 or 0.05
Higher-threshold combined	3	0.6	0.01 or 0.05

Table 2: Readable setting names. Standard factors have rank 256, four restarts and at most 25 assignment iterations. Molecular thresholds refer to the 1024-bit radius-two Morgan/Tanimoto representation. Group caps are half the smallest target fold, with the freeze rule; top-32 neighbours use exact CPU blocks in the benchmark and exact GPU tiling in the H100 scaling study. Materials grouping instead uses a representation-specific similarity quantile.

Operation	Storage or arithmetic scope
Factor storage	$O(nr)$
Factor construction/whitening	$O(nrd + nr^2 + r^3)$ arithmetic
Fold products, fixed fold count	$O(nr)$ per iteration, plus assignment sorting
M point-refinement moves	Up to $O(Mnr)$ arithmetic
Exact neighbour construction	$O(n^2 d)$ arithmetic; blocked temporary storage
Retained neighbour indices	$O(nk)$ storage

Table 3: Computational scope for n entities, feature dimension d, rank r and neighbour degree k. Input features and outputs occupy additional storage. If the number of refinement moves grows with n, refinement can be quadratic. The factor-storage statement alone is not a linear-time claim.

2 Settings and computational scope

3 Factor construction and optimization details

The main article presents the factor-space objective and its scientific use. This section records the numerical choices needed to reproduce the solver. The regularized landmark eigendecomposition floors eigenvalues at 10^{-6}.

Landmark selection affects approximation fidelity; fixed and adaptive sampling are established alternatives [Kumar et al., 2012, Musco and Musco, 2017]. The implementation supports uniform landmarks and a similarity-based spreading heuristic historically named `kmeans++`. The latter samples proportional to $(1 - \max_j S_{ij})^2$ over current landmarks. This is not ordinary squared Hilbert-distance sampling for a normalized kernel, so standard k-means++ initialization guarantees do not automatically apply. We retain its name for API compatibility and evaluate uniform sampling as an ablation.

The target count for each fold is obtained by largest-remainder rounding. For normalized requested fractions q_c, the tolerance corridor is based on $(1 \pm \epsilon)nq_c$, with integer bounds widened only enough to contain the rounded target. Zero tolerance yields the exact integer targets.

Each restart starts from a seeded count-constrained assignment. The two-fold linear subproblem is solved by sorting preference differences while enforcing both folds' bounds. Three-fold assignment uses a regret-ordered greedy peel that leaves sufficient capacity for the remaining folds. Proposals exclude self-similarity and are accepted only when the sum of squared fold sums does not decrease. A rejected proposal ends the restart. The best restart undergoes move refinement, or size-preserving pair swaps at exact capacities. The current fold is masked before alternative destinations are searched. Candidate limits make this a local search. Point-level refinement is skipped above two million entities. The standard settings are rank 256, four restarts and at most 25 assignment iterations.

For groups, the default freeze rule stops growth of the larger component after a cap-blocked merge. The alternative skip rule rejects only that merge. Weighted assignment and repair operate on group counts; groups dissolved into singletons are counted in the output diagnostics. Failure to satisfy the final count corridor raises an error. The multi-component extension concatenates per-axis factors. Dense identity columns for repeated featureless values can make its storage quadratic, so this extension is outside the reported scaling claims.

4 Neighbour construction details

The CPU reference scans exact similarities in row blocks. Blocking limits temporary storage while retaining quadratic arithmetic. Two-dimensional GPU tiling supplies a second exact backend. The optional NN-Descent candidate graph over packed binary fingerprints recomputes retained Tanimoto similarities using exact bit intersections. Its candidate recall is approximate. Top-32 recall and recall of exact top-32 edges at similarity at least 0.4 use 256 uniformly sampled query entities. Index-based recall is conservative when similarities tie. Compilation warm-up is recorded separately from warm graph construction. The retained

graph-construction measurements appear below. Their split-quality and optimizer clocks are not used to characterize the revised solver.

Entities	Backend	Graph seconds	Top-32 recall	Strong recall
1000	Exact tiled	0.009	1.000	1.000
10000	Exact tiled	0.240	1.000	1.000
100000	Exact tiled	13.002	1.000	1.000
30000	Exact tiled	1.287	1.000	1.000
1000	NN-Descent	0.842	1.000	1.000
10000	NN-Descent	8.132	0.993	1.000
100000	NN-Descent	97.569	0.990	0.998
30000	NN-Descent	26.129	0.994	1.000

Table 4: Neighbour-graph engineering measurements on ChEMBL fingerprints. Exact tiled search uses a Tesla T4; NN-Descent uses two CPU threads. Recall is measured on 256 sampled query entities; strong recall restricts exact top-32 edges to similarity at least 0.4. ANN compilation warm-up is separate. These are graph-construction observations from the preserved source study, not H100 split-call timings or measurements of the revised partition optimizer.

5 Powered-kernel fidelity and rank sensitivity

The fidelity study uses six full datasets: FreeSolv, ESOL, BACE, BBBP, ClinTox and Lipophilicity. For each power, rank and seed, a factor and optimized partition are constructed with the release solver. Five count-preserving perturbations give a spectrum of partitions evaluated against the same factor and exact powered kernel. The tables distinguish empirical cut errors from the spectral residual estimate. The latter is obtained with a symmetric eigensolver, a fixed starting vector and tolerance 10^{-8}; its eigenpair residual is retained in the machine-readable results. It is not a certified numerical upper bound.

6 External assay sensitivity

Original and class-balanced partitions share fold counts and input molecules. Class-balanced assignments use target labels, and their structural composition changes. They diagnose sensitivity and are not a proposed label-blind splitter. Per-assay JSON records retain all assignments, input and producer hashes, nearest-training-molecule similarity and label support, and all model scores. Reported ROC-AUC below 0.5 is retained; no prediction is flipped after inspecting the test labels. Class rates accompany AUC so shifts are visible directly.

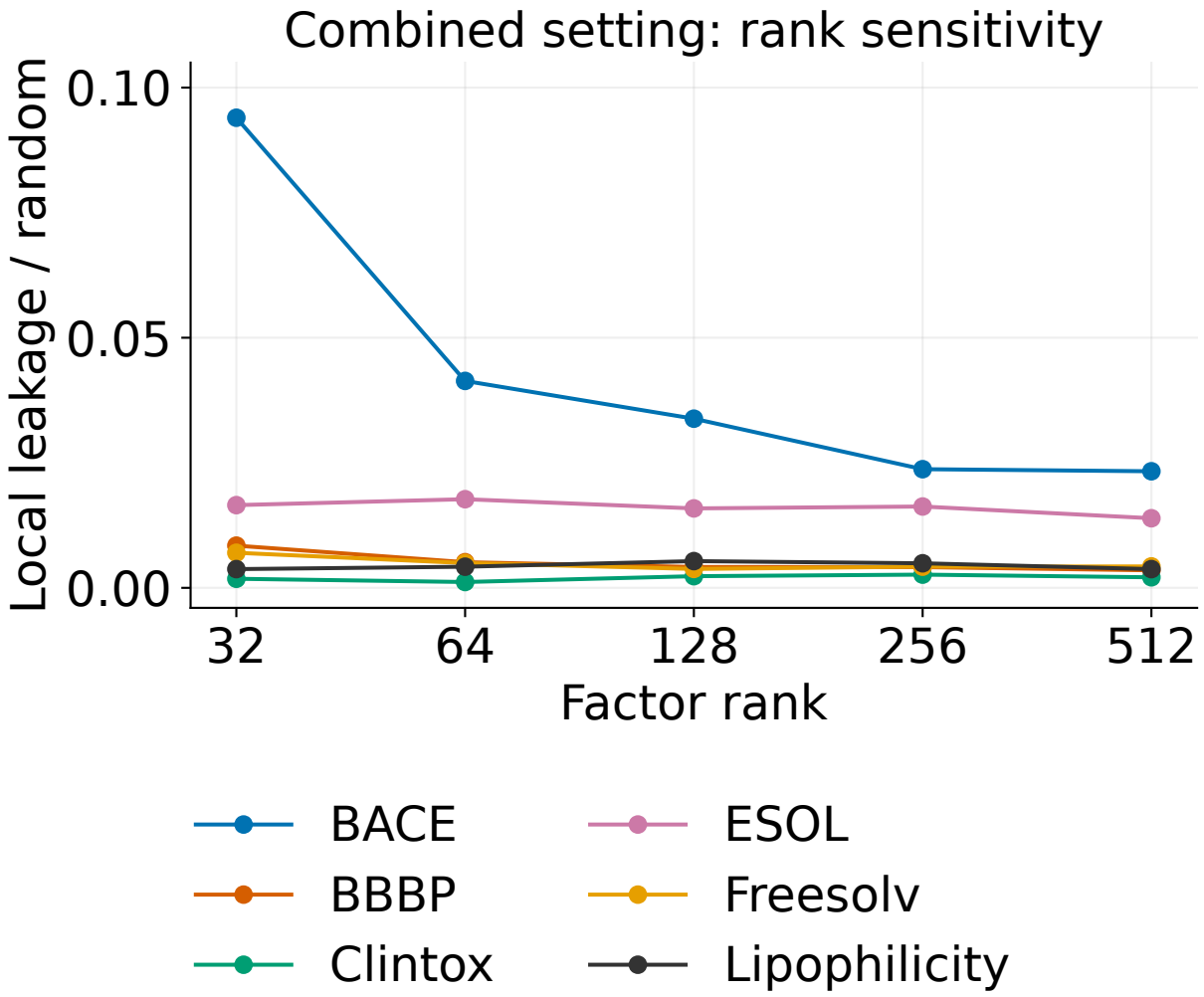


Figure 1: Combined-setting local leakage versus rank. Values average three split seeds at exact integer fold counts. Here local queries include validation and test, for continuity with the rank study's stored metric. The main baseline comparison uses test-only local queries and reports pooled held-out queries separately. Small nonmonotone changes are retained.

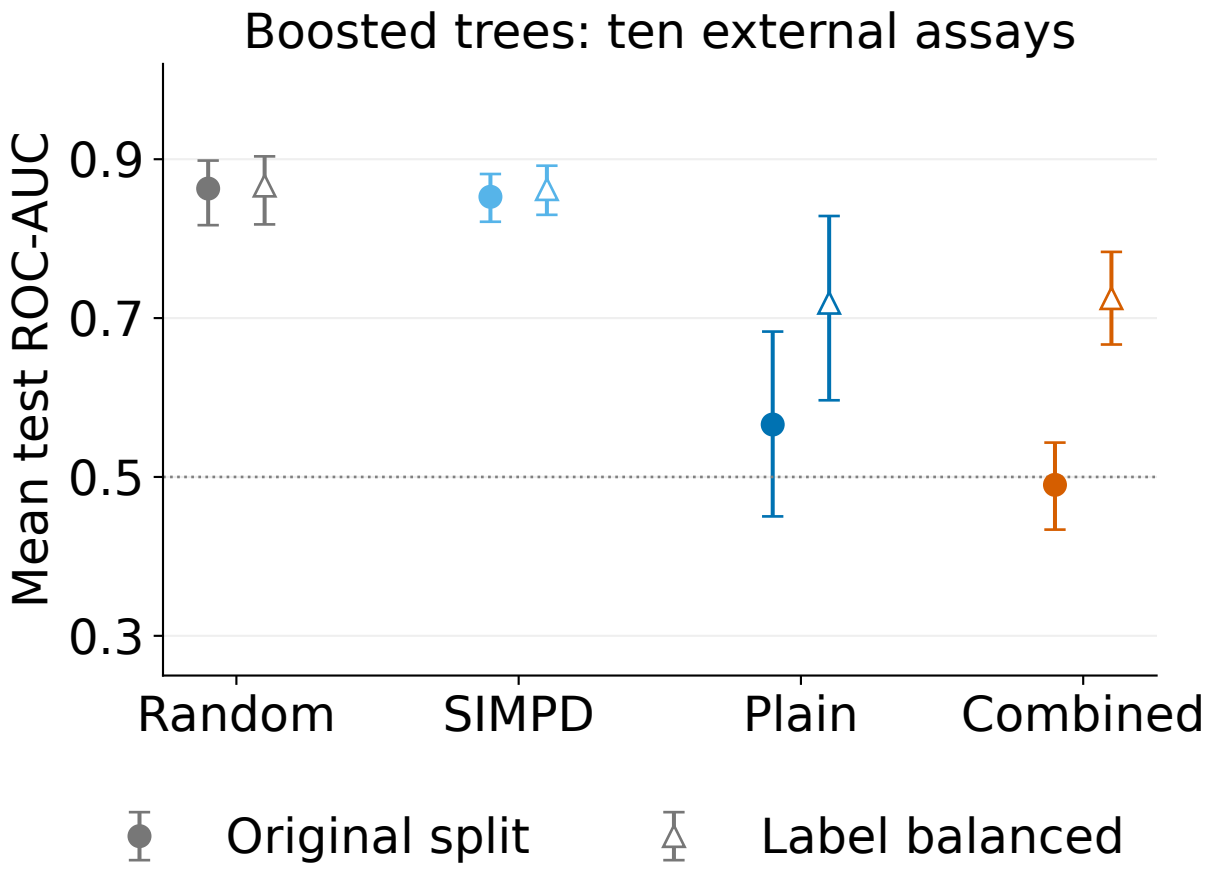


Figure 2: Boosted-tree absolute AUC with the same protocol and assay-bootstrap intervals as the main RF figure. Circles show original partitions; triangles show the target-aware intervention.

Power	Rank	Max error / mass	Median bound / mass	Max bound / mass
1	32	0.0808	0.4403	0.6789
1	64	0.0369	0.2488	0.3695
1	128	0.0164	0.1323	0.2451
1	256	0.0058	0.0716	0.1191
1	512	0.0021	0.0337	0.0569
3	32	0.3110	3.3021	7.2872
3	64	0.2788	3.1409	7.1525
3	128	0.2239	2.6336	7.0219
3	256	0.1563	2.1753	5.8603
3	512	0.0958	1.4566	2.7974

Table 5: Six full development datasets, three seeds and five partition perturbations per factor. Numerical spectral estimates use a symmetric eigensolver and are not certified upper bounds. Values above one give a vacuous bound on a cut ratio.

Assay	Random	SIMPD	Plain	Combined	Balanced
1293269-1	0.875	0.931	0.724	0.409	0.886
1889-1	0.870	0.903	0.496	0.464	0.846
225-1	0.784	0.848	0.534	0.525	0.583
237-1	0.890	0.839	0.605	0.529	0.656
269-1	0.933	0.887	0.500	0.384	0.867
2820-1	0.900	0.903	0.439	0.416	0.815
284-1	0.890	0.847	0.711	0.554	0.767
3837-1	0.932	0.901	0.456	0.410	0.545
3952-3	0.921	0.839	0.624	0.824	0.821
4105864-1	0.713	0.767	0.801	0.706	0.781

Table 6: RF mean test ROC-AUC by assay. Balanced refers to the combined partition after the target-aware intervention; it is not the label-blind method.

Assay	Random	SIMPD	Plain	Combined	Balanced
1293269-1	0.903	0.915	0.925	0.422	0.871
1889-1	0.858	0.912	0.701	0.601	0.841
225-1	0.795	0.849	0.476	0.596	0.663
237-1	0.883	0.846	0.647	0.543	0.586
269-1	0.927	0.867	0.512	0.499	0.776
2820-1	0.899	0.878	0.380	0.322	0.756
284-1	0.866	0.862	0.650	0.387	0.617
3837-1	0.909	0.861	0.515	0.560	0.620
3952-3	0.897	0.777	0.193	0.433	0.783
4105864-1	0.693	0.759	0.661	0.538	0.736

Table 7: XGB mean test ROC-AUC by assay. Balanced refers to the combined partition after the target-aware intervention; it is not the label-blind method.

Assay	Train positive rate	Test positive rate
1293269-1	0.415	0.356
1889-1	0.388	0.395
225-1	0.387	0.359
237-1	0.397	0.405
269-1	0.418	0.362
2820-1	0.403	0.396
284-1	0.412	0.390
3837-1	0.387	0.411
3952-3	0.394	0.352
4105864-1	0.389	0.451

Table 8: Class proportions for random; averages over split seeds.

Assay	Train positive rate	Test positive rate
1293269-1	0.417	0.350
1889-1	0.374	0.453
225-1	0.352	0.500
237-1	0.370	0.514
269-1	0.390	0.473
2820-1	0.391	0.444
284-1	0.388	0.486
3837-1	0.429	0.242
3952-3	0.363	0.479
4105864-1	0.410	0.367

Table 9: Class proportions for published simpd; averages over split seeds.

Assay	Train positive rate	Test positive rate
1293269-1	0.429	0.300
1889-1	0.374	0.453
225-1	0.383	0.375
237-1	0.433	0.257
269-1	0.468	0.161
2820-1	0.464	0.154
284-1	0.401	0.433
3837-1	0.386	0.414
3952-3	0.391	0.366
4105864-1	0.408	0.376

Table 10: Class proportions for plain; averages over split seeds.

Assay	Train positive rate	Test positive rate
1293269-1	0.333	0.683
1889-1	0.392	0.380
225-1	0.388	0.356
237-1	0.419	0.315
269-1	0.455	0.215
2820-1	0.466	0.145
284-1	0.400	0.438
3837-1	0.382	0.431
3952-3	0.430	0.211
4105864-1	0.430	0.287

Table 11: Class proportions for combined; averages over split seeds.

7 Scaling measurements

The plain-factor and exact-tiled combined pipelines use the same ChEMBL cache, nested sample indices, kernel rank and two-thread CPU allocation. Independent cases are distributed across reserved H100s, with one process per case. All returned assignments satisfy the stated two-sided count corridor. The table reports the actual measured ranges without extending the combined setting to unmeasured sizes. Saved sample indices and assignments allow independent checks of retention, counts and nesting.

Setting	Molecules	Seconds	GPU GiB	Host RSS GiB
Plain	1000	0.10	0.05	1.47
Plain	10000	0.32	0.16	2.12
Plain	100000	1.37	1.30	5.34
Plain	300000	3.55	3.83	11.86
Plain	1000000	10.83	12.67	34.70
Plain	2000000	19.93	25.32	67.34
Combined	1000	0.15	0.05	1.49
Combined	10000	1.67	0.16	2.13
Combined	30000	27.45	0.42	3.01
Combined	100000	89.04	1.30	5.33

Table 12: Fresh-process H100 NVL scaling cases at rank 256 and size tolerance 0.01. Split time excludes feature loading and scoring and follows a 100-entity warm-up. GPU peak is reset after warm-up; process RSS includes inputs and warm-up. The combined setting uses exact tiled neighbours. These are single-seed measurements, not repeated timing distributions.

8 Threshold and independent-representation robustness

Power/threshold cases and their controls share the same retained rows and exact 8:1:1 integer targets. Morgan and MACCS kernels are scored exactly on the full datasets, with test-only local queries and the complete training fold as reference. Three random assignments at those counts supply a common normalization for all settings within a seed. MACCS keys are never supplied to the optimizer. Its raw nearest-neighbour similarity is the primary independent representation check: the absolute Morgan ramp cutoffs have no universal interpretation for MACCS.

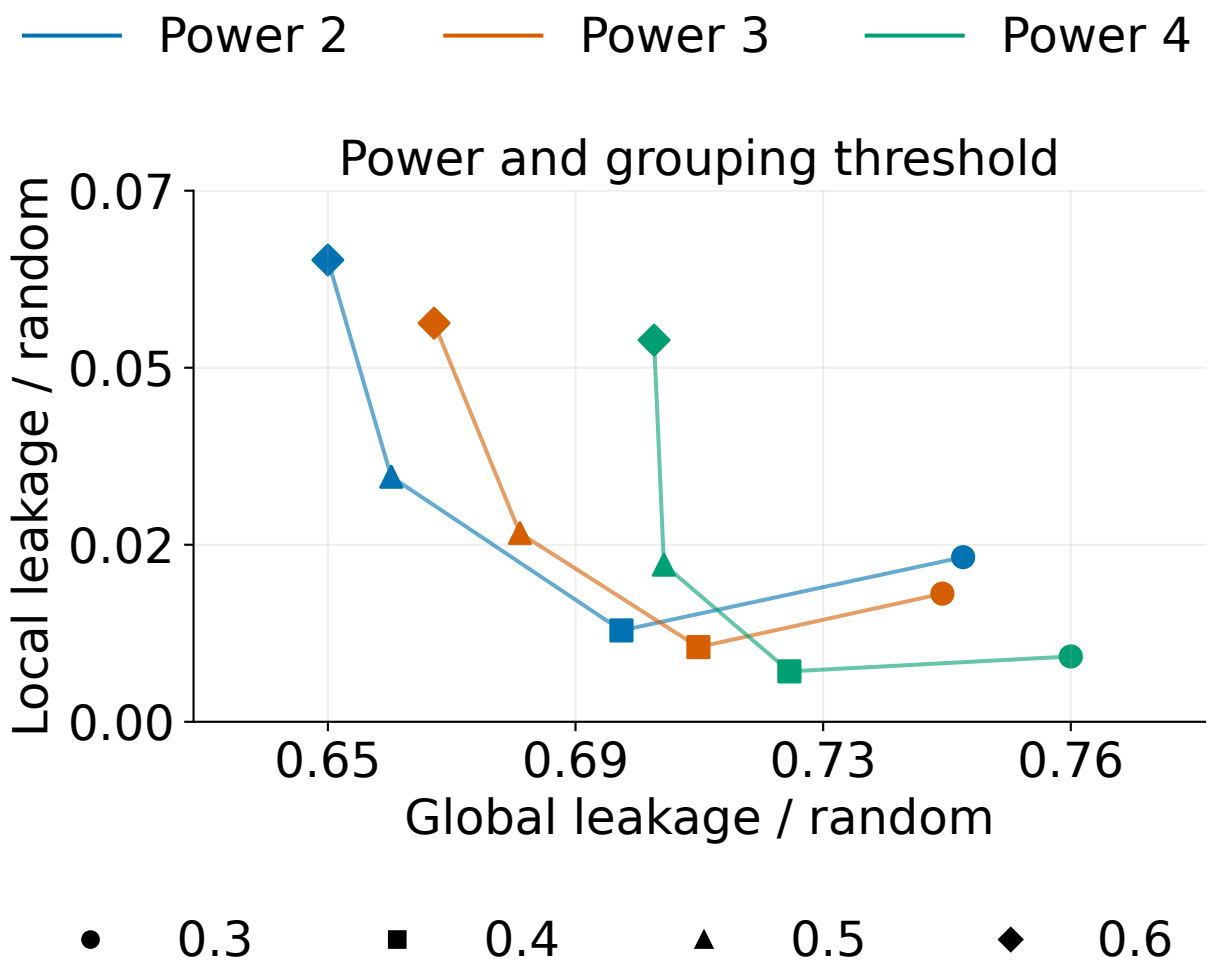


Figure 3: Power/threshold sensitivity at exact fold counts on six full development datasets. Each point averages dataset means after averaging three split seeds. Colours denote kernel power; marker shapes denote grouping thresholds, listed along the bottom. Lines connect thresholds in increasing order within each power. Both axes use the original Morgan kernel; local queries are test-only. The settings were not selected using external-assay prediction scores.

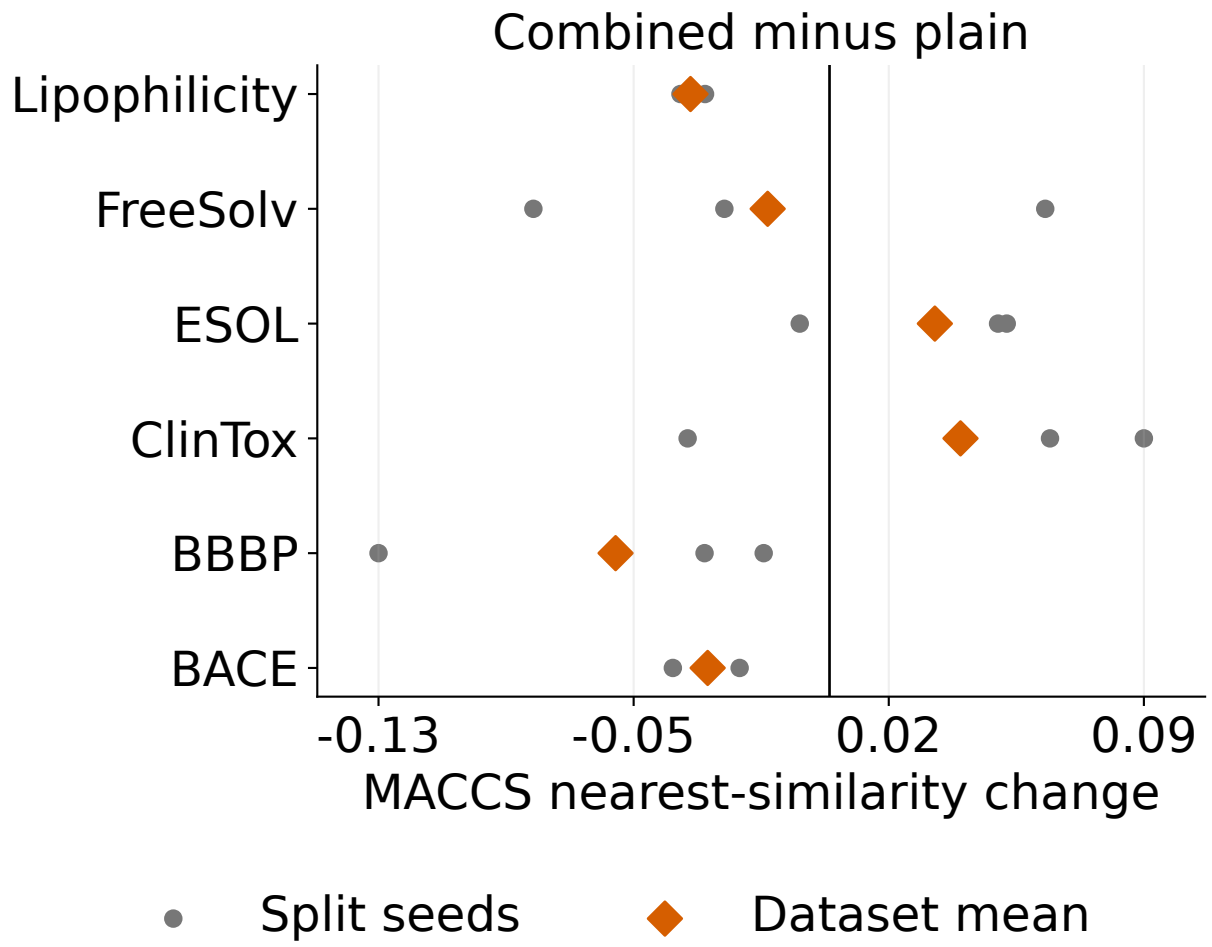


Figure 4: MACCS nearest-training-similarity changes for the fixed combined setting relative to plain low rank. Small circles are paired split-seed differences and diamonds are dataset means. Negative values indicate lower similarity. The two positive dataset means are retained.

Metric	Mean change	95% interval	Wins/ties	Sign p
Morgan ramp 0.4–0.8	-0.163	[-0.246, -0.086]	6/0	0.031
Morgan ramp 0.3–0.7	-0.201	[-0.308, -0.101]	6/0	0.031
Morgan ramp 0.5–0.9	-0.109	[-0.174, -0.054]	6/0	0.031
Morgan top-five ramp	-0.075	[-0.130, -0.020]	6/0	0.031
Morgan fraction above 0.6	-0.091	[-0.165, -0.024]	6/0	0.031
Morgan fraction above 0.8	-0.007	[-0.014, -0.001]	4/2	0.125
MACCS nearest similarity	-0.014	[-0.041, 0.015]	4/0	0.688
MACCS ramp 0.4–0.8	-0.002	[-0.060, 0.057]	4/0	0.688

Table 13: Combined minus plain on six full datasets after averaging three split seeds. Intervals are descriptive dataset bootstraps. Sign-test p-values are two-sided, exclude ties, and are unadjusted across correlated exploratory diagnostics. MACCS cutoffs are a sensitivity check, not a universal scale of chemical relatedness.

9 Component ablation

Six full datasets (BACE, BBBP, ESOL, FreeSolv, Lipophilicity and Tox21) are partitioned with the release solver at eighteen prespecified settings and three split seeds. This study separates landmark sampling, restart budget, move/swap refinement and cap policy in addition to the power, grouping, rank and capacity controls. It uses the same primary scoring protocol as the main benchmark: exact global scoring through 4,000 entities and a fixed uniform 4,000-row sample above that limit; test-only local queries against the full training fold, with three count-preserving permutation controls. All input entities are retained. The 0.05-tolerance row may change fold counts; the other rows use exact integer targets. Group dissolution counts describe output relaxations rather than a guarantee of hard analogue exclusion. NN-Descent can return fewer candidate neighbours for some samples; execution logs retain these native warnings, and the assignments are scored without tuning the search parameters after observing their results.

	Global ratio	Local ratio	Dissolved groups
Plain	0.750	0.315	–
Rank 32	0.756	0.497	–
Rank 64	0.752	0.398	–
Rank 128	0.751	0.408	–
Uniform landmarks	0.750	0.367	–
One restart	0.751	0.274	–
No move/swap refinement	0.750	0.323	–
Size tolerance 0.05	0.748	0.431	–
Power 2	0.761	0.123	–
Power 3	0.770	0.086	–
Groups only	0.777	0.149	1.000
Combined	0.786	0.022	1.000
Combined threshold 0.5	0.784	0.041	0.333
Combined threshold 0.6	0.777	0.037	0.000
Group cap 0.25	0.786	0.032	1.000
Group cap 1.0	0.789	0.013	1.000
Skip cap-blocked merges	0.791	0.240	1.333
Approximate neighbours	0.786	0.022	1.000

Table 14: bace: full release-code component ablation, averaged over three split seeds. Standard settings are rank 256, four restarts, move/swap refinement and exact counts. Combined uses power 3, threshold 0.4 and group cap 0.5. Each row changes the indicated setting; cap and neighbour rows modify Combined. Group dissolution is reported only for grouped settings. Actual fold counts and individual observations accompany the table.

	Global ratio	Local ratio	Dissolved groups
Plain	0.695	0.094	–
Rank 32	0.704	0.160	–
Rank 64	0.699	0.183	–
Rank 128	0.696	0.116	–
Uniform landmarks	0.695	0.177	–
One restart	0.696	0.126	–
No move/swap refinement	0.695	0.094	–
Size tolerance 0.05	0.685	0.112	–
Power 2	0.698	0.141	–
Power 3	0.709	0.082	–
Groups only	0.726	0.007	0.000
Combined	0.748	0.002	0.000
Combined threshold 0.5	0.715	0.021	0.000
Combined threshold 0.6	0.710	0.042	0.000
Group cap 0.25	0.734	0.000	0.000
Group cap 1.0	0.746	0.001	0.000
Skip cap-blocked merges	0.749	0.000	0.000
Approximate neighbours	0.748	0.002	0.000

Table 15: bbbp: full release-code component ablation, averaged over three split seeds. Standard settings are rank 256, four restarts, move/swap refinement and exact counts. Combined uses power 3, threshold 0.4 and group cap 0.5. Each row changes the indicated setting; cap and neighbour rows modify Combined. Group dissolution is reported only for grouped settings. Actual fold counts and individual observations accompany the table.

	Global ratio	Local ratio	Dissolved groups
Plain	0.528	0.099	–
Rank 32	0.532	0.146	–
Rank 64	0.530	0.154	–
Rank 128	0.529	0.151	–
Uniform landmarks	0.528	0.117	–
One restart	0.528	0.109	–
No move/swap refinement	0.528	0.087	–
Size tolerance 0.05	0.519	0.089	–
Power 2	0.548	0.128	–
Power 3	0.608	0.069	–
Groups only	0.553	0.023	0.000
Combined	0.644	0.023	0.000
Combined threshold 0.5	0.604	0.041	0.000
Combined threshold 0.6	0.579	0.075	0.000
Group cap 0.25	0.643	0.022	0.000
Group cap 1.0	0.631	0.013	0.667
Skip cap-blocked merges	0.646	0.021	0.000
Approximate neighbours	0.644	0.023	0.000

Table 16: esol: full release-code component ablation, averaged over three split seeds. Standard settings are rank 256, four restarts, move/swap refinement and exact counts. Combined uses power 3, threshold 0.4 and group cap 0.5. Each row changes the indicated setting; cap and neighbour rows modify Combined. Group dissolution is reported only for grouped settings. Actual fold counts and individual observations accompany the table.

	Global ratio	Local ratio	Dissolved groups
Plain	0.420	0.122	–
Rank 32	0.429	0.193	–
Rank 64	0.423	0.284	–
Rank 128	0.420	0.140	–
Uniform landmarks	0.420	0.228	–
One restart	0.420	0.154	–
No move/swap refinement	0.420	0.123	–
Size tolerance 0.05	0.406	0.011	–
Power 2	0.431	0.079	–
Power 3	0.519	0.056	–
Groups only	0.501	0.004	0.000
Combined	0.505	0.006	0.333
Combined threshold 0.5	0.485	0.010	0.000
Combined threshold 0.6	0.455	0.051	0.000
Group cap 0.25	0.502	0.006	0.000
Group cap 1.0	0.506	0.010	0.000
Skip cap-blocked merges	0.510	0.002	0.000
Approximate neighbours	0.505	0.009	0.000

Table 17: freesolv: full release-code component ablation, averaged over three split seeds. Standard settings are rank 256, four restarts, move/swap refinement and exact counts. Combined uses power 3, threshold 0.4 and group cap 0.5. Each row changes the indicated setting; cap and neighbour rows modify Combined. Group dissolution is reported only for grouped settings. Actual fold counts and individual observations accompany the table.

	Global ratio	Local ratio	Dissolved groups
Plain	0.801	0.335	–
Rank 32	0.811	0.359	–
Rank 64	0.805	0.423	–
Rank 128	0.803	0.320	–
Uniform landmarks	0.801	0.298	–
One restart	0.801	0.317	–
No move/swap refinement	0.802	0.319	–
Size tolerance 0.05	0.796	0.307	–
Power 2	0.804	0.300	–
Power 3	0.809	0.272	–
Groups only	0.865	0.004	0.000
Combined	0.871	0.004	0.000
Combined threshold 0.5	0.820	0.019	0.000
Combined threshold 0.6	0.815	0.060	0.000
Group cap 0.25	0.845	0.003	0.000
Group cap 1.0	0.872	0.003	0.000
Skip cap-blocked merges	0.875	0.002	0.000
Approximate neighbours	0.871	0.004	0.000

Table 18: lipophilicity: full release-code component ablation, averaged over three split seeds. Standard settings are rank 256, four restarts, move/swap refinement and exact counts. Combined uses power 3, threshold 0.4 and group cap 0.5. Each row changes the indicated setting; cap and neighbour rows modify Combined. Group dissolution is reported only for grouped settings. Actual fold counts and individual observations accompany the table.

	Global ratio	Local ratio	Dissolved groups
Plain	0.573	0.210	–
Rank 32	0.579	0.299	–
Rank 64	0.577	0.182	–
Rank 128	0.572	0.152	–
Uniform landmarks	0.574	0.172	–
One restart	0.573	0.151	–
No move/swap refinement	0.573	0.217	–
Size tolerance 0.05	0.565	0.175	–
Power 2	0.588	0.264	–
Power 3	0.607	0.250	–
Groups only	0.655	0.105	0.000
Combined	0.721	0.025	0.000
Combined threshold 0.5	0.651	0.045	0.000
Combined threshold 0.6	0.623	0.116	0.000
Group cap 0.25	0.719	0.023	0.000
Group cap 1.0	0.698	0.023	0.000
Skip cap-blocked merges	0.715	0.023	0.000
Approximate neighbours	0.721	0.025	0.000

Table 19: tox21: full release-code component ablation, averaged over three split seeds. Standard settings are rank 256, four restarts, move/swap refinement and exact counts. Combined uses power 3, threshold 0.4 and group cap 0.5. Each row changes the indicated setting; cap and neighbour rows modify Combined. Group dissolution is reported only for grouped settings. Actual fold counts and individual observations accompany the table.

10 Published local and cluster baselines

Lo-Hi uses the official Hi objective, threshold 0.4 and no coarsening. Resource controls cap it at two CPU threads and 7,200 solver seconds, with the native solver outcome retained. Eligible full datasets have at most 5,000 molecules. The comparison preserves the native retained subset and exact realized counts; discarded molecules are never assigned to a test fold by another name.

The UMAP recipe follows the published NCI-60 clustering study [Hernández-Hernández and Ballester, 2023]: radius-two 1024-bit Morgan fingerprints, Jaccard UMAP with 100 neighbours, minimum distance zero, two components and seed 42, followed by Ward agglomeration into seven clusters. We transfer this recipe to new datasets and evaluate all seven native cluster holdouts. Low-rank controls exactly match each holdout's counts and retain all rows. The memory policy excludes cases whose condensed float64 agglomeration distances alone exceed 16 GiB. FreeSolv instead exhibits a native method failure: one valid fingerprint is disconnected in Jaccard similarity, producing a nonfinite embedding coordinate rejected by Ward clustering. We neither impute nor drop the row to manufacture a result.

Dataset	Plain G	Plain N	Combined G	Combined N	Lo-Hi G	Lo-Hi N
bace	0.729	0.314	0.764	0.004	0.948	0.000
bbbp	0.607	0.126	0.640	0.000	0.988	0.000
clintox	0.562	0.162	0.595	0.000	0.932	0.000
esol	0.432	0.107	0.533	0.012	0.933	0.000
freesolv	0.309	0.128	0.412	0.000	0.801	0.000
lipophilicity	0.756	0.318	0.805	0.003	0.974	0.000
sider	0.466	0.117	0.494	0.000	0.989	0.000

Table 20: Lo-Hi comparisons use each native retained subset and its exact train/test counts. Low-rank settings average three seeds. Retained fractions and native solver states are reported separately. G and N denote global and local ratios, evaluated within each study protocol.

Dataset	Plain G	Plain N	Combined G	Combined N	UMAP G	UMAP N
bace	0.678	0.224	0.726	0.047	0.863	0.034
bbbp	0.605	0.124	0.638	0.000	0.923	0.012
clintox	0.585	0.142	0.635	0.001	0.922	0.042
esol	0.448	0.108	0.546	0.016	0.843	0.063
hiv	0.581	0.223	0.715	0.023	0.983	0.109
lipophilicity	0.757	0.295	0.810	0.003	0.951	0.027
qm8	0.573	0.315	0.684	0.043	0.893	0.053
sider	0.466	0.125	0.495	0.000	0.913	0.031
tox21	0.503	0.228	0.621	0.025	0.888	0.143

Table 21: UMAP comparisons average all seven native cluster holdouts equally within each dataset. Low-rank controls retain all entities and exactly match each holdout count. FreeSolv is an audited native embedding/clustering failure; MUV exceeds the prespecified memory policy. Neither is assigned a score. G and N denote global and local ratios, evaluated within each study protocol.

11 Graph-degree and optimization sensitivity

Graph degree is varied over 16, 32, 64 and 128 on six full development datasets with three seeds and tolerance 0.01. The prespecified primary comparator uses 32 neighbours, matching the grouping builder's degree. All settings use the ordinary pairwise Graph API, quality preset and CUT objective. Larger graphs retain more dense similarity mass but change the global–local trade-off; no single degree is declared best for both metrics. Native and repaired assignments are preserved. Tables distinguish degree sensitivity from the additional effect of imposing both lower and upper capacity bounds.

The FreeSolv search control compares four and thirty restarts at exact counts for plain and combined settings, with five fixed seeds. A small change under this finite search budget is empirical saturation, not a lower bound on the unknown optimum. The combined setting shows larger objective changes than the plain setting, so the earlier development claim that extra restarts have no effect is not carried forward.

Neighbours	Retained mass	Global ratio	Local ratio
16	0.0624	0.8218	0.0138
32	0.1057	0.8103	0.0184
64	0.1749	0.7639	0.0538
128	0.2824	0.7166	0.0296

Table 22: Graph degree sensitivity on six full datasets. Dataset means average three seeds before cross-dataset averaging. Retained mass uses quantized graph weights and the full ordinary-kernel mass, with quantization-error bounds supplied in the CSV. Global and local scores follow the main common protocol. Build, partition and capacity repair are included in split time.

Setting	Dense cut change	Powered cut change	Local ratio change
combined	-0.0182	-0.0297	0.0214
plain	-0.0004	-0.0004	-0.0687

Table 23: Thirty versus four restarts on full FreeSolv at exact counts, averaged over five fixed seeds. Cut changes are relative to the four-restart value; local changes are absolute. Both use the same factor seed within each pair. More restarts do not certify optimality, and choosing a different pre-refinement basin can change the final polished solution.

12 Materials generality check

Leaving related materials together is a recognized evaluation choice [Meredig et al., 2018, Witman and Schindler, 2025], and structure-based out-of-distribution benchmarks test transfer beyond random holdouts [Omee et al., 2024]. Here the QMOF and Materials Project checks use composition descriptors and a standardized Euclidean similarity kernel. They test applicability of the factor formulation beyond molecular Tanimoto similarity, rather than crystal-structure novelty. Full per-dataset scores accompany the benchmark tables. The study makes no empirical claim about untested sequence or biomolecular featurizers.

Dataset	Plain G	Plain N	Combined G	Combined N	DataSAIL G	DataSAIL N
bace	0.750	0.363	0.785	0.066	0.775	0.007
bbbp	0.692	0.099	0.743	0.001	0.731	0.002
clintox	0.658	0.082	0.698	0.001	0.699	0.020
esol	0.524	0.093	0.635	0.027	0.597	0.072
freesolv	0.415	0.357	0.497	0.009	0.510	0.000
lipophilicity	0.799	0.314	0.869	0.004	0.956	0.013
sider	0.594	0.091	0.625	0.000	0.634	0.000

Table 24: Exact-count controls for saved DataSAIL assignments at nominal tolerance 0.01. Each low-rank partition retains the same entities and has exactly the three native DataSAIL counts. Low-rank scores average three seeds; DataSAIL has one assignment per configuration. G and N denote global and test-only local ratios. These reused assignments supply no new runtime.

Recorded outcome	Tolerance 0.01	Tolerance 0.05
feasible assignment; no optimality certificate	3	3
no feasible assignment returned within configured solve	1	1
no recorded attempt under this protocol	3	3
resource-policy exclusion	2	2
solver-reported optimum of clustered problem	4	4

Table 25: DataSAIL source coverage, classified from native logs and saved assignments. Solver-reported optima concern its clustered integer program. The legacy CSV label "infeasible" is not treated as proof of infeasibility when no such solver certificate was retained. Resource exclusions and missing recorded attempts are separate.

13 Calibration protocol and prediction provenance

The hardness parameter uses a nested seeded sequence of point swaps between an optimized assignment and its count-preserving random endpoint. Its observed predictive response need not be monotone. Classification difficulty is $\mathrm{AUC}_0 - \mathrm{AUC}_\alpha$; regression difficulty is $\mathrm{RMSE}_\alpha/\mathrm{RMSE}_0 - 1$. Subscript zero denotes the random endpoint with the same split seed,

fold counts and model. These definitions use test scores and avoid interpreting a nearly perfect RF training score as a separate measure of difficulty.

Five knots, $\alpha \in \{0, 0.25, 0.5, 0.75, 1\}$, are fitted with RF on split seeds 0–2. The mean curve uses isotonic regression followed by piecewise-linear interpolation. Its direction is estimated from the calibration probes. Fitting can pool noisy negative early differences, so the fitted random endpoint can differ slightly from the known observed value zero. Seeds 10–14 are excluded from that fit and evaluate all five knots, as well as three interior targets at 25%, 50% and 75% of the fitted difficulty range. RF and boosted trees each have model initialization seed zero. The study has 570 model fits and 378 saved assignments across six datasets. Observed adjacent-knot decreases are reported before isotonic fitting; a fitted monotone curve is not evidence that individual predictive responses are monotone.

Fixed-knot validation and interior-target inversion evaluate different points. Their errors therefore receive separate tables and source columns. The random-relative study's authoritative summary is `calibration_v2/validation_summary.csv`. The older `calibration_validation.csv` and `inversion_validation.csv` describe 25 fixed-knot and 15 interior-target scores per dataset/model, respectively, using train-minus-test gaps. Recomputed legacy errors match both files; they are separate studies and are not used as random-relative calibration errors. The source reconciliation is included with the release records.

Dataset	Units	RF MAE	XGB MAE
bace	AUC decrement	0.041	0.023
bbbp	AUC decrement	0.052	0.047
esol	relative RMSE	0.092	0.096
freesolv	relative RMSE	0.555	0.624
lipophilicity	relative RMSE	0.044	0.083
sider	AUC decrement	0.030	0.042

Table 26: Fixed-knot validation on five unseen split seeds (25 points per model and dataset). Classification uses AUC(random) minus AUC(split); regression uses RMSE(split)/RMSE(random) minus one. No test label is used to fit the calibration on validation seeds. Each model has initialization seed zero.

Dataset	Units	RF MAE	XGB MAE
bace	AUC decrement	0.056	0.030
bbbp	AUC decrement	0.062	0.054
esol	relative RMSE	0.101	0.115
freesolv	relative RMSE	0.711	0.747
lipophilicity	relative RMSE	0.060	0.096
sider	AUC decrement	0.067	0.069

Table 27: Interior-target inversion on the same five validation seeds (15 points per model and dataset). Targets are fitted using RF calibration seeds 0–2. Classification uses AUC(random) minus AUC(split); regression uses RMSE(split)/RMSE(random) minus one. No test label is used to fit the calibration on validation seeds. Each model has initialization seed zero.

Dataset	Model	Decreases	Adjacent steps
bace	rf	6	20
bace	xgb	3	20
bbbp	rf	6	20
bbbp	xgb	5	20
esol	rf	8	20
esol	xgb	4	20
freesolv	rf	4	20
freesolv	xgb	5	20
lipophilicity	rf	0	20
lipophilicity	xgb	1	20
sider	rf	4	20
sider	xgb	8	20

Table 28: Observed decreases in random-relative difficulty as hardness increases on validation seeds. Four adjacent knot steps are checked for each of five seeds. Isotonic fitting does not remove these violations from the observed scores.

The D-MPNN metric-validation study reuses saved predictions from the specified 30-epoch protocol and five model initializations. Test SMILES are checked against the prediction rows before ensembling. Its split assignments and prior group-cap rule remain attached to those measurements. It is a supporting metric study, separate from the release-code thirteen-dataset benchmark. Per-model and per-split variability is reported at its actual level of replication.

14 Metric-validation provenance

The synthetic analysis excludes random-reference and all hypergraph rows. Global and local correlations use the same finite observations. Correlations by constructed regime and all included rows are distributed as CSV files; pooled associations can also reflect differences between regimes.

Per-molecule validation uses existing five-initialization D-MPNN predictions for the random and plain-low-rank partitions of four datasets. Prediction rows are checked against test SMILES before ensembling. Training-only references are primary. Training-plus-validation references are retained as a secondary check. Within-split correlations accompany pooled results because the latter also mix split membership with molecular similarity. These analyses validate metric interpretation; they are not new release-code neural-model experiments.

Regime	Observations	Global rho	Local rho
continuum	24	0.844	0.578
hierarchy	30	0.402	0.491
long_tail	30	-0.022	0.631
near_dup	24	-0.290	0.588
pooled	120	0.020	0.519
tightness	12	0.435	0.808

Table 29: Family-leakage correlations on identical finite observations for both metrics. Local similarity is more strongly associated in four of the five constructed regimes; the continuum regime favors the global measure. The pooled contrast also reflects differences among regimes.

Dataset	Partition	Test entities	NN rho	Global rho
BBBP	Plain	387	0.064	0.209
BBBP	Random	408	-0.078	-0.142
ESOL	Plain	214	-0.189	-0.074
ESOL	Random	226	-0.094	0.078
FreeSolv	Plain	121	-0.375	-0.072
FreeSolv	Random	128	-0.109	-0.042
Lipophilicity	Plain	798	-0.187	-0.058
Lipophilicity	Random	840	-0.217	0.039

Table 30: Within-partition correlations with absolute five-initialization D-MPNN ensemble error. NN similarity uses the actual training fold as reference; the global proxy is the per-point overlap score. Each partition has one split realization. Pooled and training-plus-validation analyses accompany these primary within-partition values as CSV files.

15 Complete benchmark tables

Nominal size tolerance 0.01

	Global ratio	Local ratio	Split seconds
Random	1.003	0.914	0.009
Plain	0.420	0.123	0.132
Combined	0.504	0.009	0.187
Higher threshold	0.455	0.051	0.235
k-NN graph	0.547	0.026	0.324
Scaffold	0.760	0.240	0.052
Butina	0.826	0.261	0.097
DataSAIL	0.510	0.000	–

Table 31: freesolv at tolerance 0.01. Metrics average split seeds; time is the median split-call time in the accompanying row records. DataSAIL assignments are reused and have no new time. Dashes mean unavailable under the stated protocol. Actual counts, group relaxations, raw timing observations and all exclusion reasons are retained in the CSV files.

	Global ratio	Local ratio	Split seconds
Random	1.005	1.013	0.013
Plain	0.526	0.087	0.169
Combined	0.641	0.021	0.305
Higher threshold	0.581	0.068	0.379
k-NN graph	0.727	0.018	0.410
Scaffold	0.929	0.469	0.156
Butina	0.881	0.215	0.319
DataSAIL	0.597	0.072	–

Table 32: esol at tolerance 0.01. Metrics average split seeds; time is the median split-call time in the accompanying row records. DataSAIL assignments are reused and have no new time. Dashes mean unavailable under the stated protocol. Actual counts, group relaxations, raw timing observations and all exclusion reasons are retained in the CSV files.

Nominal size tolerance 0.05

	Global ratio	Local ratio	Split seconds
Random	0.994	0.977	0.012
Plain	0.750	0.373	0.207
Combined	0.784	0.016	0.360
Higher threshold	0.776	0.035	0.317
k-NN graph	0.918	0.035	0.410
Scaffold	0.981	0.865	0.621
Butina	0.861	0.346	0.764
DataSAIL	0.775	0.007	–

Table 33: bace at tolerance 0.01. Metrics average split seeds; time is the median split-call time in the accompanying row records. DataSAIL assignments are reused and have no new time. Dashes mean unavailable under the stated protocol. Actual counts, group relaxations, raw timing observations and all exclusion reasons are retained in the CSV files.

	Global ratio	Local ratio	Split seconds
Random	1.000	1.011	0.015
Plain	0.693	0.152	0.266
Combined	0.744	0.003	0.591
Higher threshold	0.707	0.038	0.558
k-NN graph	0.887	0.011	0.533
Scaffold	1.013	0.517	0.577
Butina	0.924	0.142	1.103
DataSAIL	0.731	0.002	–

Table 34: bbbp at tolerance 0.01. Metrics average split seeds; time is the median split-call time in the accompanying row records. DataSAIL assignments are reused and have no new time. Dashes mean unavailable under the stated protocol. Actual counts, group relaxations, raw timing observations and all exclusion reasons are retained in the CSV files.

	Global ratio	Local ratio	Split seconds
Random	1.001	0.996	0.011
Plain	0.596	0.084	0.206
Combined	0.623	0.000	0.425
Higher threshold	0.615	0.040	0.453
k-NN graph	0.793	0.007	0.463
Scaffold	1.037	0.564	2.103
Butina	0.934	0.126	0.643
DataSAIL	0.634	0.000	–

Table 35: sider at tolerance 0.01. Metrics average split seeds; time is the median split-call time in the accompanying row records. DataSAIL assignments are reused and have no new time. Dashes mean unavailable under the stated protocol. Actual counts, group relaxations, raw timing observations and all exclusion reasons are retained in the CSV files.

	Global ratio	Local ratio	Split seconds
Random	0.999	0.959	0.010
Plain	0.660	0.071	0.199
Combined	0.698	0.001	0.435
Higher threshold	0.682	0.063	0.439
k-NN graph	0.834	0.003	0.476
Scaffold	1.022	0.527	0.507
Butina	0.931	0.117	0.626
DataSAIL	0.699	0.020	–

Table 36: clintox at tolerance 0.01. Metrics average split seeds; time is the median split-call time in the accompanying row records. DataSAIL assignments are reused and have no new time. Dashes mean unavailable under the stated protocol. Actual counts, group relaxations, raw timing observations and all exclusion reasons are retained in the CSV files.

	Global ratio	Local ratio	Split seconds
Random	1.001	1.006	0.033
Plain	0.802	0.318	0.447
Combined	0.869	0.004	1.788
Higher threshold	0.814	0.062	1.284
k-NN graph	0.948	0.017	1.027
Scaffold	0.983	0.651	1.380
Butina	0.945	0.111	4.740
DataSAIL	0.956	0.013	–

Table 37: lipophilicity at tolerance 0.01. Metrics average split seeds; time is the median split-call time in the accompanying row records. DataSAIL assignments are reused and have no new time. Dashes mean unavailable under the stated protocol. Actual counts, group relaxations, raw timing observations and all exclusion reasons are retained in the CSV files.

	Global ratio	Local ratio	Split seconds
Random	1.016	1.003	0.075
Plain	0.571	0.218	0.774
Combined	0.722	0.025	4.120
Higher threshold	0.623	0.117	3.507
k-NN graph	0.750	0.018	2.462
Scaffold	0.959	0.454	1.582
Butina	0.920	0.235	16.415
DataSAIL	–	–	–

Table 38: tox21 at tolerance 0.01. Metrics average split seeds; time is the median split-call time in the accompanying row records. DataSAIL assignments are reused and have no new time. Dashes mean unavailable under the stated protocol. Actual counts, group relaxations, raw timing observations and all exclusion reasons are retained in the CSV files.

	Global ratio	Local ratio	Split seconds
Random	1.002	1.008	0.329
Plain	0.640	0.595	3.673
Combined	0.703	0.012	17.568
Higher threshold	0.670	0.120	40.124
k-NN graph	0.686	0.006	13.593
Scaffold	0.941	0.603	2.029
Butina	–	–	–
DataSAIL	–	–	–

Table 39: qm8 at tolerance 0.01. Metrics average split seeds; time is the median split-call time in the accompanying row records. DataSAIL assignments are reused and have no new time. Dashes mean unavailable under the stated protocol. Actual counts, group relaxations, raw timing observations and all exclusion reasons are retained in the CSV files.

	Global ratio	Local ratio	Split seconds
Random	0.953	0.996	223.143
Plain	0.688	0.205	13.679
Combined	0.832	0.029	70.735
Higher threshold	0.706	0.124	98.239
k-NN graph	0.879	0.051	41.908
Scaffold	0.987	0.573	12.942
Butina	–	–	–
DataSAIL	–	–	–

Table 40: hiv at tolerance 0.01. Metrics average split seeds; time is the median split-call time in the accompanying row records. DataSAIL assignments are reused and have no new time. Dashes mean unavailable under the stated protocol. Actual counts, group relaxations, raw timing observations and all exclusion reasons are retained in the CSV files.

	Global ratio	Local ratio	Split seconds
Random	0.979	0.999	1.330
Plain	0.779	0.516	14.229
Combined	0.907	0.256	261.990
Higher threshold	0.840	0.288	974.538
k-NN graph	0.949	0.109	364.558
Scaffold	0.972	0.748	32.771
Butina	–	–	–
DataSAIL	–	–	–

Table 41: muv at tolerance 0.01. Metrics average split seeds; time is the median split-call time in the accompanying row records. DataSAIL assignments are reused and have no new time. Dashes mean unavailable under the stated protocol. Actual counts, group relaxations, raw timing observations and all exclusion reasons are retained in the CSV files.

	Global ratio	Local ratio	Split seconds
Random	0.991	1.003	0.075
Plain	0.735	0.179	1.572
Combined	0.761	0.028	10.763
Higher threshold	0.761	0.028	11.139
k-NN graph	0.818	0.005	5.665
Scaffold	–	–	–
Butina	–	–	–
DataSAIL	–	–	–

Table 42: qmof at tolerance 0.01. Metrics average split seeds; time is the median split-call time in the accompanying row records. DataSAIL assignments are reused and have no new time. Dashes mean unavailable under the stated protocol. Actual counts, group relaxations, raw timing observations and all exclusion reasons are retained in the CSV files.

	Global ratio	Local ratio	Split seconds
Random	0.991	0.997	0.660
Plain	0.783	0.299	7.136
Combined	0.819	0.144	129.969
Higher threshold	0.819	0.144	125.466
k-NN graph	0.869	0.192	71.073
Scaffold	–	–	–
Butina	–	–	–
DataSAIL	–	–	–

Table 43: materials project at tolerance 0.01. Metrics average split seeds; time is the median split-call time in the accompanying row records. DataSAIL assignments are reused and have no new time. Dashes mean unavailable under the stated protocol. Actual counts, group relaxations, raw timing observations and all exclusion reasons are retained in the CSV files.

	Global ratio	Local ratio	Split seconds
Random	1.003	0.914	0.005
Plain	0.406	0.011	0.134
Combined	0.490	0.009	0.188
Higher threshold	0.440	0.040	0.248
k-NN graph	0.547	0.027	0.335
Scaffold	0.760	0.240	0.052
Butina	0.826	0.261	0.097
DataSAIL	0.494	0.023	–

Table 44: freesolv at tolerance 0.05. Metrics average split seeds; time is the median split-call time in the accompanying row records. DataSAIL assignments are reused and have no new time. Dashes mean unavailable under the stated protocol. Actual counts, group relaxations, raw timing observations and all exclusion reasons are retained in the CSV files.

	Global ratio	Local ratio	Split seconds
Random	1.005	1.013	0.012
Plain	0.519	0.089	0.170
Combined	0.626	0.024	0.299
Higher threshold	0.572	0.077	0.362
k-NN graph	0.668	0.022	0.381
Scaffold	0.929	0.469	0.160
Butina	0.881	0.215	0.319
DataSAIL	0.588	0.073	–

Table 45: esol at tolerance 0.05. Metrics average split seeds; time is the median split-call time in the accompanying row records. DataSAIL assignments are reused and have no new time. Dashes mean unavailable under the stated protocol. Actual counts, group relaxations, raw timing observations and all exclusion reasons are retained in the CSV files.

	Global ratio	Local ratio	Split seconds
Random	0.994	0.977	0.011
Plain	0.748	0.431	0.215
Combined	0.776	0.008	0.343
Higher threshold	0.771	0.021	0.321
k-NN graph	0.915	0.042	0.417
Scaffold	0.981	0.865	0.617
Butina	0.861	0.346	0.754
DataSAIL	0.763	0.012	–

Table 46: bace at tolerance 0.05. Metrics average split seeds; time is the median split-call time in the accompanying row records. DataSAIL assignments are reused and have no new time. Dashes mean unavailable under the stated protocol. Actual counts, group relaxations, raw timing observations and all exclusion reasons are retained in the CSV files.

	Global ratio	Local ratio	Split seconds
Random	1.000	1.011	0.015
Plain	0.685	0.112	0.248
Combined	0.736	0.001	0.593
Higher threshold	0.700	0.038	0.561
k-NN graph	0.887	0.010	0.575
Scaffold	1.013	0.517	0.571
Butina	0.924	0.142	1.075
DataSAIL	0.722	0.002	–

Table 47: bbbp at tolerance 0.05. Metrics average split seeds; time is the median split-call time in the accompanying row records. DataSAIL assignments are reused and have no new time. Dashes mean unavailable under the stated protocol. Actual counts, group relaxations, raw timing observations and all exclusion reasons are retained in the CSV files.

	Global ratio	Local ratio	Split seconds
Random	1.001	0.996	0.011
Plain	0.584	0.110	0.205
Combined	0.611	0.000	0.370
Higher threshold	0.605	0.050	0.419
k-NN graph	0.797	0.008	0.493
Scaffold	1.037	0.564	2.096
Butina	0.934	0.126	0.656
DataSAIL	0.620	0.000	–

Table 48: sider at tolerance 0.05. Metrics average split seeds; time is the median split-call time in the accompanying row records. DataSAIL assignments are reused and have no new time. Dashes mean unavailable under the stated protocol. Actual counts, group relaxations, raw timing observations and all exclusion reasons are retained in the CSV files.

	Global ratio	Local ratio	Split seconds
Random	0.999	0.959	0.011
Plain	0.650	0.074	0.200
Combined	0.688	0.001	0.402
Higher threshold	0.673	0.057	0.467
k-NN graph	0.837	0.021	0.499
Scaffold	1.022	0.527	0.491
Butina	0.931	0.117	0.633
DataSAIL	0.690	0.021	–

Table 49: clintox at tolerance 0.05. Metrics average split seeds; time is the median split-call time in the accompanying row records. DataSAIL assignments are reused and have no new time. Dashes mean unavailable under the stated protocol. Actual counts, group relaxations, raw timing observations and all exclusion reasons are retained in the CSV files.

	Global ratio	Local ratio	Split seconds
Random	1.001	1.006	0.034
Plain	0.796	0.307	0.458
Combined	0.868	0.004	1.711
Higher threshold	0.810	0.072	1.269
k-NN graph	0.918	0.025	1.032
Scaffold	0.983	0.651	1.370
Butina	0.945	0.111	4.701
DataSAIL	0.985	0.010	–

Table 50: lipophilicity at tolerance 0.05. Metrics average split seeds; time is the median split-call time in the accompanying row records. DataSAIL assignments are reused and have no new time. Dashes mean unavailable under the stated protocol. Actual counts, group relaxations, raw timing observations and all exclusion reasons are retained in the CSV files.

	Global ratio	Local ratio	Split seconds
Random	1.016	1.003	0.078
Plain	0.565	0.175	0.873
Combined	0.710	0.025	3.902
Higher threshold	0.616	0.116	3.685
k-NN graph	0.747	0.024	2.376
Scaffold	0.959	0.454	1.590
Butina	0.920	0.235	16.277
DataSAIL	–	–	–

Table 51: tox21 at tolerance 0.05. Metrics average split seeds; time is the median split-call time in the accompanying row records. DataSAIL assignments are reused and have no new time. Dashes mean unavailable under the stated protocol. Actual counts, group relaxations, raw timing observations and all exclusion reasons are retained in the CSV files.

	Global ratio	Local ratio	Split seconds
Random	1.002	1.008	0.321
Plain	0.636	0.618	2.929
Combined	0.799	0.083	544.601
Higher threshold	0.669	0.130	31.234
k-NN graph	0.681	0.009	13.487
Scaffold	0.941	0.603	2.013
Butina	–	–	–
DataSAIL	–	–	–

Table 52: qm8 at tolerance 0.05. Metrics average split seeds; time is the median split-call time in the accompanying row records. DataSAIL assignments are reused and have no new time. Dashes mean unavailable under the stated protocol. Actual counts, group relaxations, raw timing observations and all exclusion reasons are retained in the CSV files.

	Global ratio	Local ratio	Split seconds
Random	0.953	0.996	5.400
Plain	0.685	0.243	6.900
Combined	0.830	0.033	75.497
Higher threshold	0.701	0.139	102.224
k-NN graph	0.853	0.048	44.556
Scaffold	0.987	0.573	12.903
Butina	–	–	–
DataSAIL	–	–	–

Table 53: hiv at tolerance 0.05. Metrics average split seeds; time is the median split-call time in the accompanying row records. DataSAIL assignments are reused and have no new time. Dashes mean unavailable under the stated protocol. Actual counts, group relaxations, raw timing observations and all exclusion reasons are retained in the CSV files.

	Global ratio	Local ratio	Split seconds
Random	0.979	0.999	1.286
Plain	0.774	0.515	21.295
Combined	0.915	0.238	243.311
Higher threshold	0.833	0.290	1075.031
k-NN graph	0.949	0.111	359.014
Scaffold	0.972	0.748	40.553
Butina	–	–	–
DataSAIL	–	–	–

Table 54: muv at tolerance 0.05. Metrics average split seeds; time is the median split-call time in the accompanying row records. DataSAIL assignments are reused and have no new time. Dashes mean unavailable under the stated protocol. Actual counts, group relaxations, raw timing observations and all exclusion reasons are retained in the CSV files.

	Global ratio	Local ratio	Split seconds
Random	0.991	1.003	0.055
Plain	0.729	0.182	1.425
Combined	0.752	0.028	12.158
Higher threshold	0.752	0.028	12.082
k-NN graph	0.786	0.007	6.095
Scaffold	–	–	–
Butina	–	–	–
DataSAIL	–	–	–

Table 55: qmof at tolerance 0.05. Metrics average split seeds; time is the median split-call time in the accompanying row records. DataSAIL assignments are reused and have no new time. Dashes mean unavailable under the stated protocol. Actual counts, group relaxations, raw timing observations and all exclusion reasons are retained in the CSV files.

	Global ratio	Local ratio	Split seconds
Random	0.991	0.997	0.246
Plain	0.781	0.281	7.325
Combined	0.810	0.130	121.645
Higher threshold	0.810	0.130	122.091
k-NN graph	0.915	0.187	75.454
Scaffold	–	–	–
Butina	–	–	–
DataSAIL	–	–	–

Table 56: materials project at tolerance 0.05. Metrics average split seeds; time is the median split-call time in the accompanying row records. DataSAIL assignments are reused and have no new time. Dashes mean unavailable under the stated protocol. Actual counts, group relaxations, raw timing observations and all exclusion reasons are retained in the CSV files.

Comparison	Metric	n	Mean change	95% interval	Holm p
Plain / k-NN graph	Global	11	-0.163	[-0.18, -0.13]	0.020
Plain / k-NN graph	Local	11	0.222	[0.13, 0.32]	0.020
Combined / Plain	Global	11	0.082	[0.06, 0.11]	0.020
Combined / Plain	Local	11	-0.215	[-0.31, -0.14]	0.020
Combined / k-NN graph	Global	11	-0.081	[-0.11, -0.05]	0.152
Combined / k-NN graph	Local	11	0.007	[-0.01, 0.04]	1.000
Plain / DataSAIL	Global	7	-0.067	[-0.10, -0.04]	0.156
Plain / DataSAIL	Local	7	0.184	[0.09, 0.28]	0.156
Combined / DataSAIL	Global	7	-0.007	[-0.04, 0.02]	1.000
Combined / DataSAIL	Local	7	-0.001	[-0.02, 0.02]	1.000

Table 57: Paired molecular-dataset effects at tolerance 0.01. Changes subtract the second method from the first. Dataset-bootstrap intervals use 10,000 resamples after averaging split seeds. Sign-test p values receive Holm correction across the 20 declared comparisons (five method pairs, two metrics and two tolerances). DataSAIL pairs use exact-count controls. These convenience datasets support descriptive, representation-specific inference.

16 Same-host timing details

Dataset	Method	Median seconds	Global ratio	Local ratio
bace	combined	0.555	0.785	0.066
bace	datasail	7210.760	0.766	0.019
bace	plain	0.279	0.750	0.363
bbbp	combined	1.041	0.743	0.001
bbbp	datasail	7219.393	0.731	0.002
bbbp	plain	0.337	0.692	0.099
clintox	combined	1.057	0.698	0.001
clintox	datasail	1411.427	0.699	0.020
clintox	plain	0.434	0.658	0.082
lipophilicity	combined	2.700	0.869	0.004
lipophilicity	datasail	8796.726	0.956	0.013
lipophilicity	plain	1.095	0.799	0.314

Table 58: Same-host split-call time and exact-count quality. DataSAIL distance construction, clustering and integer programming are inside its clock; imports, feature loading and scoring are outside. Low-rank calls exclude a 100-entity warm-up. DataSAIL has one solve per dataset; low-rank entries aggregate three seeds. The solver time limit is 7,200 seconds and is not an end-to-end wall-time bound. Native solver states and all three low-rank observations accompany the table.

A Objective identities and conditional bounds

Let $H \in \{0,1\}^{n\times k}$ be the assignment matrix, with one nonzero per row, and let h_c denote its columns. The cross-fold mask is $\mathbf{1}\mathbf{1}^\top - HH^\top$ and has zero diagonal. Consequently,

$$\mathcal{C}_S(H) = \tfrac{1}{2}\left(\mathbf{1}^\top S\mathbf{1} - \sum_c h_c^\top S h_c\right). \tag{1}$$

Substituting $S = BB^\top$ yields Equation 2. Group contraction is exact because a whole group's contribution to every block sum is the sum of its rows. This statement concerns retained groups; dissolving a group changes the admissible partition set rather than invalidating the identity.

For a positive-semidefinite kernel with features ϕ_i, its integer elementwise power has features $\phi_i^{\otimes p}$ because $\langle \phi_i^{\otimes p}, \phi_j^{\otimes p}\rangle = \langle \phi_i, \phi_j\rangle^p$. This gives the stated positive-semidefinite property directly and explains why arbitrary fractional powers are not accepted.

For nonnegative similarities, let $N_\tau(z)$ count unordered crossing pairs with $S_{ij} \geq \tau > 0$. Each such pair contributes at least τ^p, so

$$\mathcal{C}_{S^{\circ p}}(z) \geq \tau^p N_\tau(z). \tag{2}$$

Thus the powered cut supplies a soft penalty on the number of strong crossing pairs. For a fixed nonempty set of crossing pairs, $\lim_{p\to\infty} \mathcal{C}_{S^{\circ p}}(z)^{1/p} = \max_{i<j:z_i\neq z_j} S_{ij}$, by the finite-dimensional ℓ_p norm limit. This is the maximum over all crossing pairs, not the average of per-test-point nearest neighbours used in NNL. The power therefore provides a principled tail surrogate, without making the optimized objective identical to the evaluation metric. A rank-limited factor introduces a separate approximation error for each power.

For symmetric residual $E = S - BB^\top$, the spectral norm inequality $|v^\top Ev| \leq \|E\|_2\|v\|^2$ gives

$$|\mathcal{C}_S(H) - \mathcal{C}_{BB^\top}(H)| \leq \tfrac{1}{2}\|E\|_2 \left(\|\mathbf{1}\|^2 + \sum_c \|h_c\|^2 \right) \tag{3}$$

$$= n\|E\|_2 \equiv \delta. \tag{4}$$

The bound is uniform across assignments and can be loose in practice. If a solver returns $\widehat{H}$ with approximate-kernel objective at most η above the minimum over a specified feasible set, then

$$\mathcal{C}_S(\widehat{H}) \leq \min_{H \text{ feasible}} \mathcal{C}_S(H) + 2\delta + \eta. \tag{5}$$

Our heuristic does not certify η, so this conditional inequality is not an empirical optimality guarantee. It also requires comparing the same feasible set; grouping, count changes and kernel sharpening alter the optimization problem.

Moving a factor row b from block a to c replaces (p_a, p_c) by $(p_a - b, p_c + b)$. Expansion of Equation 2 gives $\Delta\mathcal{C} = b^\top p_a - b^\top p_c - \|b\|^2$. For a swap of factor rows u and v, the reduction equals the sum of their single-move reductions minus $2u^\top v$. These identities hold for group sums as well. Searching only a finite set of swap candidates preserves the validity of an accepted move, but does not establish that no improving swap exists elsewhere.

If all fold counts equal a fixed value m, kernel k-means maximizes $m^{-1}\sum_c \|p_c\|^2$ and agrees with dense-cut minimization. For unequal fixed counts it maximizes a differently weighted sum, and for a capacity corridor the weights themselves change with the assignment. This is why we do not transfer ordinary kernel k-means convergence or approximation guarantees to the present solver without further assumptions.

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